

How the saturating law emerges from the effective-distance ↔ correlation-correction trade-off

$$\alpha^*(p) = \frac{L}{1 + (p_0/p)^m}$$

Summary. The law is the minimal interpolation between two asymptotic regimes forced by two competing effects that live at *different orders* in the physical error rate p . The plateau L is the internal optimum of the correlation-correction (benefit) sector; the tail exponent m is set by the geometry of the spurious detours that cost correlated matching one unit of fault-tolerant distance (harm); and the crossover p_0 is where the two become comparable. Each fitted parameter maps onto a mechanistic quantity, verified against the sweep CSVs and the exhaustive fault enumeration.

1. Bookkeeping: the two effects live at different fault orders

Expand the logical error rate as a fault-weight series at distance d , with $n = \lceil d/2 \rceil$ ($n = 3$ for $d = 5$):

$$\text{LER}(\alpha, p) = [C - B(\alpha)] p^n + H(\alpha, p) p^{n-1}$$

- **Benefit sector (correlation correction).** Genuine Y- and CNOT-correlated faults live in the *same* p^n sector as MWPM's failures — correlated matching lowers the *coefficient* of p^n by tipping near-degenerate matchings the right way. $B(\alpha)$ rises from 0 and has an internal maximum at $\alpha = A \approx 0.75 < 1$: even within the p^n sector, full strength over-tips some decisions.

- **Harm sector (effective-distance loss).** The spurious-detour failures need one fault *fewer* — this is the enumeration result: 337 weight-2 chains defeat $\text{CM}(\alpha=1)$ at $d = 5$, whereas MWPM needs 3 faults to fail. So the harm enters one order down, at p^{n-1} .

The structural key is that **harm / benefit** $\sim 1/p$: the two sectors are not in fixed proportion — the harm's relative weight grows without bound as $p \rightarrow 0$. Were the overcorrections to cost at the *same* fault order as the benefit, the trade-off would be p -independent and α^* would be a constant. The entire p -dependence of α^* exists *because* the harm lives one order below the benefit.

2. The α -dependence of each sector

Harm is a threshold staircase that slides with p . A spurious detour built from k boosted edges (each costing $k(|\log \alpha| + c)$ after damping) beats an honest route of margin $\Delta \approx a \cdot \log(1/p)$ if and only if

$$k(|\log \alpha| + c) < \Delta \approx a \log(1/p) \Leftrightarrow \alpha > \alpha_{\text{th}} \propto p^{a/k}.$$

So $H(\alpha, p)$ is a sum over configurations of (probability coefficient) $\times \theta(\alpha > \alpha_{\text{th}})$ — a staircase in α whose steps were measured directly (failing pairs $2 \rightarrow 6 \rightarrow 14 \rightarrow 54 \rightarrow 154 \rightarrow 225 \rightarrow 373$ as α goes $0.05 \rightarrow 1$ at $p = 2 \times 10^{-4}$), and whose thresholds all slide toward 0 as $p^{a/k}$. The measured shift is $\approx \sqrt{10}$ per decade, i.e. $a/k \approx 1/2$.

Benefit is p -independent in α . $B(\alpha)$ is a property of the correlation structure (which decisions need tipping, and by how much), not of the ambient weight scale, so its optimum

A does not move with p . Empirical check: the fitted plateau is $L = 0.775$ ($d = 5$) and 0.749 ($d = 7$) — the *same* number at two distances, as required if it is the benefit sector's internal optimum rather than anything tied to the harm.

3. First-order condition and the two regimes

Minimising LER over α gives

$$B'(\alpha^*) p^n = \partial_\alpha H(\alpha^*, p) p^{n-1} \Leftrightarrow B'(\alpha^*) = \frac{1}{p} \partial_\alpha H(\alpha^*, p).$$

Regime 1 — $p \gg p_0$ (near threshold). The threshold cloud $\alpha_{\text{th}} \propto \sqrt{p}$ sits *above* the benefit optimum A: at the α you would want for benefit reasons, no harmful configurations are active ($\partial H \approx 0$). The condition reduces to $B'(\alpha^*) = 0$, so

$$\alpha^* \rightarrow A \approx 0.75$$

(plateau: the harm sector is simply out of reach)

This is also why the literature, benchmarking near threshold, always found $\alpha \approx 1$ correlated matching to be fine.

Regime 2 — $p \ll p_0$. The cloud slides below A. Sitting at A would now activate a large part of the p^{n-1} -amplified staircase, and the $1/p$ factor makes ∂H overwhelming, so the optimum is *pinned to the edge of the moving threshold cloud* (plus a fixed $O(1)$ offset into it, where the marginal benefit tail balances the first few staircase steps — which is why the swept $\alpha^* = 0.2$ sits slightly above the strict onset $\alpha_{\text{th}} \approx 0.05$ – 0.1). Since every threshold scales as $p^{a/k}$,

$$\alpha^* \propto p^{a/k} \approx \sqrt{p}$$

(power-law tail)

Check against data: the swept α^* falls $0.4 \rightarrow 0.1$ from $p = 10^{-3}$ to 10^{-4} , a factor ≈ 4 per decade $\approx p^{0.6}$ — consistent with $\frac{1}{2}$ within the grid resolution, and independently confirmed by the α_{th} scaling from the enumeration.

4. The saturating law is the minimal interpolation of these asymptotics

Any function with $\lim_{p \gg p_0} \alpha^* = A$ and $\lim_{p \ll p_0} \alpha^* = A(p/p_0)^m$ is, at lowest order, exactly the saturating law, with every parameter now identified:

fit parameter	mechanistic meaning	check
plateau L = A ≈ 0.75	internal optimum of the benefit sector (p - and d -independent)	same value at $d = 5$ and $d = 7$ ✓
tail m = a/k ≈ ½	margin-to-boost-count geometry of the dominant spurious detours	α_{th} map shifts $\sqrt{10}$ per decade ✓
crossover $p_0 \approx 7 \times 10^{-4}$	where the distance-loss channel at $\alpha = A$ becomes comparable to the benefit-sector stakes	where CM(1)/MWPM peels away from best- α /MWPM in the ratio table ✓

5. Two honest caveats

- The staircase is discrete and broad, and the optimum is *shallow* (the LER is flat to $\sim 3\%$ over $\alpha \in [0.15, 0.4]$ at $p = 4 \times 10^{-4}$): the “law” describes the bottom of a wide valley, not a sharp point. The least-squares m from the fit (1.4 at $d = 5$) is dragged upward by plateau-region points and should not be read as the true tail exponent — the tail itself says $\approx \frac{1}{2}$.
- The exponent $a/k \approx \frac{1}{2}$ is currently *measured*, not derived from geometry. Deriving it means characterising the 337 failing pairs (how many boosted edges k the detour uses vs the honest margin in units of $\log(1/p)$); the enumeration map already contains the full threshold distribution $N(\alpha_{\text{th}} < \alpha)$, so this last step is extractable.

In one sentence

The plateau is the correlation-correction optimum, the tail is the effective-distance-restoration constraint, and the saturating law is nothing but the crossover between a p -independent benefit and a harm whose relative weight grows as $1/p$ while its activation thresholds slide as $\sqrt{p}$.